

ANSWER KEY
HMT28-A

1. b) 0
2. b) $30t - 12$
3. b) $y = C_1 e^{t/2} + C_2 e^{-3t/2}$
4. b) $y = e^{-t} [A \cos 2t + B \sin 2t]$
5. c) $(C_1 + C_2 \ln t) t^{-5/2}$
6. b) $C_1 e^t + C_2 e^{-t} - 2t^2 - 4$
7. b) $e^{-bt/a}$
8. c) -2
9. b) $y = e^{\alpha t} [A \cos \beta t + B \sin \beta t]$
10. Cancelled
11. a) $y(x-2) = \frac{x^3}{3} - 4x + c$
12. a) $y = \frac{-\sin 3x}{9} + e^x + \frac{x^2}{12} + \frac{1}{3}x - 1$
13. c) $C_1 e^{\left(\frac{1+\sqrt{2}}{3}\right)x} + C_2 e^{\left(\frac{1-\sqrt{2}}{3}\right)x}$
14. b) $(C_1 + C_2 x) e^{5x}$
15. Cancelled
16. c) $C_1 e^{-x} + C_2 e^{2x} + \frac{1}{2} e^{3x}$
17. b) $C_1 t^2 + C_2 t^{-4}$
18. b) $y = C_1 t^2 + C_2 t^3$
19. a) $y = C_1 t^3 + C_2 t^3 \ln |t|$
20. b) Euler equation
21. d) $g(t) = 3t e^{2t} + C e^{2t}$
22. a) $y = e^{2x} [A \cos x + B \sin x]$
23. b) $C_1 e^{-x} + C_2 e^{-2x}$
24. d) ഇവയൊന്നുമല്ല
25. a) $y = C_1 e^{-x} + C_2 x e^{-x}$